

KARTHEEK PAMIDIMUKKALA

SENIOR SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineer with 10 years designing, building, and operating large-scale distributed systems and enterprise platforms. Currently lead AI-native internal apps and enterprise platforms at Meta, designing reusable multi-agent workflows with tool routing, structured outputs, and human approvals, RAG pipelines, MCP connectors, eval harnesses, and NL to SQL data access that turn ambiguous business workflows across Finance and operations into reliable internal apps. Strong in Python backend engineering, system design, enterprise integrations (auth, idempotency, retries, rate-limit handling), and data architecture including ingestion, quality checks, lineage, and governance. Track record of building reusable platform components, observability, and high-performance services (300% throughput gains) adopted across teams.

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Meta

Apr 2022 to Present

- Lead the Tax Data and Revenue Platform serving Finance teams; designed core data architecture including high-throughput ingestion pipelines, target schemas, quality checks, data lineage tracking, and governance controls, and built a natural language to SQL interface enabling non-technical stakeholders to safely self-serve data.
- Led the design and delivery of an AI tax-audit agent platform, an internal app that drafts comprehensive responses with structured outputs using RAG over historical audit archives, integrated human-in-the-loop approvals and full audit trails, reducing audit response times by 90%.
- Built secure agent tool interfaces using an MCP-style architecture with tool routing, granting governed access to data and business logic with strict permission and safety boundaries, improving agent response times by 60%.
- Led the eval and feedback-loop infrastructure for the audit agent: eval harness, regression tests, monitoring, and metrics, raising response accuracy to 90%.
- Integrated third-party tax engines and internal finance systems via fault-tolerant APIs with idempotency keys, exponential backoff, and rate-limit handling, ensuring exactly-once processing of revenue transactions.
- Designed and implemented high-performance caching services improving application response times by 80%.
- Implemented audit trails and role-based permissions for all agent-initiated actions on financial data, meeting internal compliance and SOX audit requirements.
- Partnered with system owners and business owners to turn messy, ambiguous enterprise workflows into reliable internal apps adopted across Finance and operations.

Software Engineer, Angi Inc

Dec 2021 to Apr 2022

- Improved customer conversion flows by 30%.
- Designed software solutions by analyzing system performance standards, increasing performance efficiency by 20%.

Senior Software Engineer, Bandwidth Inc

Jul 2021 to Dec 2021

- Created IaC templates for deploying scalable microservices to AWS saving 60% of developer time.
- Created guidelines for monitoring and logging for the entire platform.

Cloud Platform Engineer, Lenovo

Dec 2019 to Jul 2021

- Solved cross-region database sync problem required for certain business critical flows.
- Architected and optimized high concurrency backend services to handle extreme traffic scaling, achieving a 300% throughput gain from 30K to 100K concurrent connections.
- Created dashboard for monitoring the performance of 30 services and provided a standard template across teams for creating dashboards for their services.
- Acted as mentor for junior engineers and lead small engineering teams.

Software Engineer, Citrix Systems

Sept 2016 to Dec 2019

- Redesigned legacy monolithic infrastructure into decoupled microservices, leveraging robust system design principles to reduce production error rates by 90%.

- Mitigated DDoS attack on the back-end systems by implementing rate limiting.

EDUCATION

Texas A&M University

Jan 2015 to May 2016

Masters in Computer Science

JNTUH, Hyderabad

Sept 2010 to Apr 2014

Bachelors in Computer Science

AREAS OF EXPERTISE

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|----------------|--------------|-------------------|-----------------------|
| • Python | • RAG | • LLM | • Agent Orchestration |
| • Apache Kafka | • Git | • SQL | • Java |
| • C# | • Amazon AWS | • Microsoft Azure | • Spring Boot |
| • AngularJS | • Docker | • Kubernetes | • C++ |
| • FastAPI | • Flask | • MongoDB | • Redis |
| • DynamoDB | • MCP | • Evals | • Structured Outputs |